SINGAPORE UNIVERSITY OF SOCIAL SCIENCES

ENG201

Examination - July Semester 2023

Linear Systems Analysis and Design

Tuesday, 21 November 2023 **04:00 pm - 06:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **FOUR (04)** questions and comprises **FIVE (05)** printed pages (including cover page).
2. You must answer **ALL** questions.
3. All answers must be written in the answer book.
4. This is **Open-book** examination.

At the end of the examination.
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.
If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

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Answer all questions. (100 marks)

Question 1

(a) Determine the fundamental period of the signal

$$x(t) = 2\cos(10t + 1) - \sin(4t - 1)$$

(3 marks)

(b) Determine the fundamental period of the signal $x[n] = 1 + e^{j\frac{4\pi n}{7}} - e^{j\frac{2\pi n}{5}}$.

(3 marks)

(c) Consider a continuous-time system with input $x(t)$ and output $y(t)$ related by equation $y(t) = x(\sin(t))$.

(i) Analyze if this system is causal.

(2 marks)

(ii) Analyze if this system is linear.

(3 marks)

(d) A continuous-time signal $x(t)$ is shown in Figure Q1(d). Sketch and label carefully each of the following signals

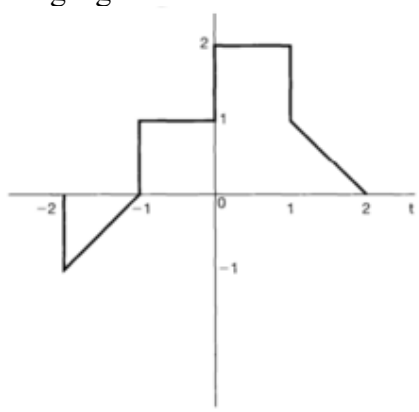

Figure Q1(d)

(i) $x(2t + 1)$

(3 marks)

(ii) $x\left(4 - \frac{t}{2}\right)$

(3 marks)

(e) Consider a system S with input $x[n]$ and output, $y[n]$. This system is obtained through a series interconnection of a system S_1 followed by a system S_2. The input-output relationship for S_1 and S_2 are:

$$S_1: \quad y_1[n] = 2x_1[n] + 4x_1[n-1]$$
$$S_2: \quad y_2[n] = x_2[n-2] + \frac{1}{2}x_2[n-3]$$

where $x_1[n]$ and $x_2[n]$ denote the input signals.

(i) Determine the input-output relationship for system S.

(4 marks)

(ii) Does the input-output relationship of system S change if the order in which S_1 and S_2, that are connected in series, is reversed?

(4 marks)

Question 2

(a) The response of a linear time-invariant system to an input signal $x(t)$ is the output signal $y(t)$. By differentiating the convolution integral, show that the response of the system to the input

$$x'(t) = \frac{dx(t)}{dt}$$

is given by

$$y'(t) = \frac{dy(t)}{dt}.$$

(8 marks)

(b) A linear time-invariant system has the response given by:

$$y(t) = \sin(\omega_0 t)$$

to the input

$$x(t) = e^{-5t}u(t).$$

Use the result obtained in Question 2(a) to describe the impulse response of this system.

(10 marks)

(c) Solve the problem in Question 2(b) using only Laplace Transforms and verify your result.

(7 marks)

Question 3

(a) Calculate the continuous-time Fourier transform for each of the following periodic signals:

(i) $x(t) = \sin\left(2\pi t + \frac{\pi}{4}\right)$

(5 marks)

(ii) $x(t) = 1 + \cos\left(6\pi t + \frac{\pi}{8}\right)$

(5 marks)

(b) Consider a linear time-invariant (LTI) System with the following impulse response:

$$h(t) = \frac{\sin(8\pi(t-1))}{\pi(t-1)}$$

Describe the response of the system for each of the following excitations:

(i) $x_1(t) = \cos\left(12\pi t + \frac{\pi}{2}\right)$

(3 marks)

(ii) $x_2(t) = \sum_{k=0}^{\infty}\left(\frac{1}{2}\right)^k \sin(6\pi k t)$

(4 marks)

(iii) $x_3(t) = \frac{\sin(8\pi(t+1))}{\pi(t+1)}$

(3 marks)

(c) Consider a causal linear time-invariant system described by the following difference equation:

$$y[n] + \frac{1}{2}y[n-1] = x[n].$$

(i) Determine the frequency response $H(e^{j\Omega})$ of the system.

(3 marks)

(ii) Determine the impulse response $h[n]$ of the system.

(2 marks)

Question 4

(a) A continuous-time linear time-invariant system with input $x(t)$ and output $y(t)$ is governed by the following differential equation:

$$\frac{d^2y(t)}{dx^2(t)} + 5\frac{dy(t)}{dx(t)} + 6y(t) = x(t)$$

for $t \geq 0$.

(i) Describe the transfer function in the Laplace domain.

(3 marks)

(ii) Sketch the pole-zero diagram in the Laplace domain. What are the pole location(s)? What are the zero location(s)?

(3 marks)

(iii) Determine the impulse response of the system.

(3 marks)

(iv) Describe the step response of the system in the time domain.

(3 marks)

(b) A discrete-time system having the transfer function in the z-domain

$$H(z) = \frac{1}{z^{-2} + 3z^{-1} - 10},$$

is excited by a finite number of sequence $\{1, 2, 3, 4, 0, 0, \}$. Assuming the system be at rest prior to the application of the input sequence:

(i) Calculate the output sequence.

(6 marks)

(ii) Discuss the stability of the discrete-time system.

(3 marks)

(iii) Sketch a block diagram for the realization of $H(z)$.

(4 marks)

----- END OF PAPER -----

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